Research Article
MASLD and Alcohol-Related Liver Diseases

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Genetic risk accentuates dietary effects on hepatic steatosis, inflammation and fibrosis in a population-based cohort

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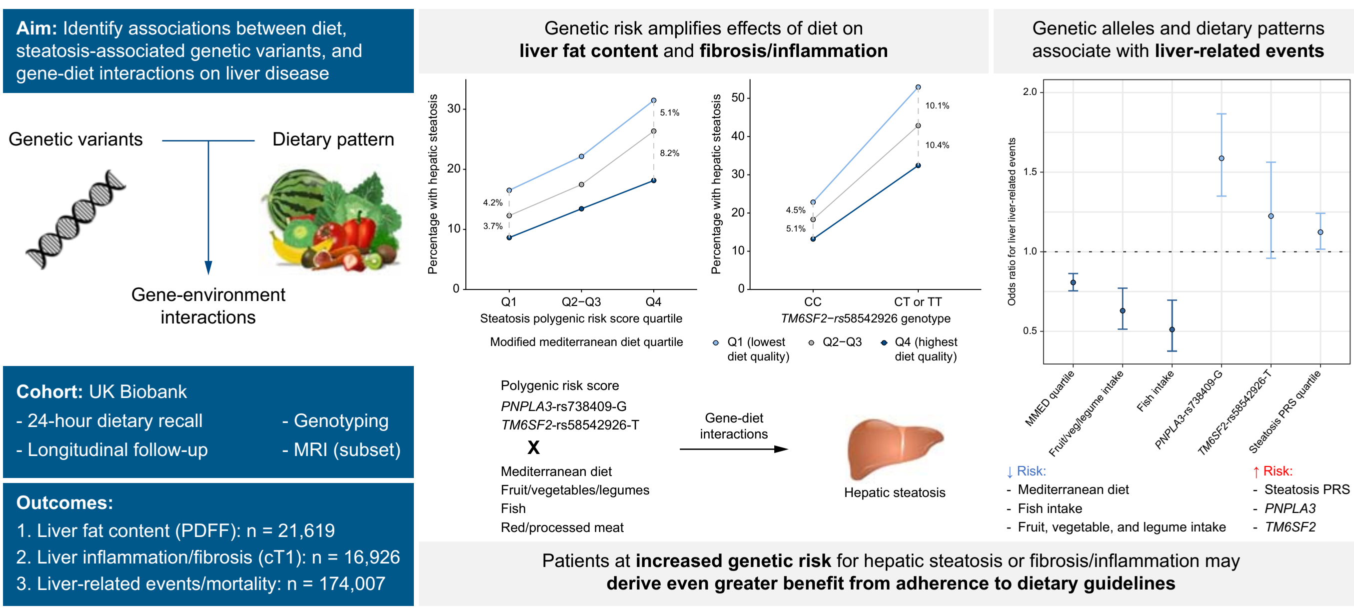
Graphical abstract



Highlights

- Poor diet quality and specific genetic alleles are associated with steatotic liver disease.
- Diet has a much greater impact on hepatic steatosis in those at higher genetic risk.
- These effects were seen for both liver fat content and iron-controlled T1 time (a measure of fibrosis and inflammation).
- Hence, dietary interventions may have a greater impact in populations at higher genetic risk.

Impact and implications

Genetic variants and diet both influence risk of hepatic steatosis, inflammation/fibrosis, and hepatic decompensation; however, how gene-diet interactions influence these outcomes has previously not been comprehensively characterized. We investigated this topic in the community-based UK Biobank and found that genetic risk and dietary quality interacted to influence hepatic steatosis and inflammation/fibrosis on liver MRI, so that the effects of diet were greater in people at elevated genetic risk. These results are relevant for patients and medical providers because they show that genetic risk is not fixed (*i.e.* modifiable factors can mitigate or exacerbate this risk) and realistic dietary changes may result in meaningful improvement in liver steatosis and inflammation/fibrosis. As genotyping becomes more routinely used in clinical practice, patients identified to be at high baseline genetic risk may benefit even more from intensive dietary counseling than those at lower risk, though future prospective studies are required.

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Genetic risk accentuates dietary effects on hepatic steatosis, inflammation and fibrosis in a population-based cohort

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Background & Aims: Steatotic liver disease (SLD), characterized by elevated liver fat content (LFC), is influenced by genetics and diet. However, whether diet has a differential effect based on genetic risk is not well-characterized. We aimed to determine how genetic factors interact with diet to affect SLD in a large national biobank.

Methods: We included UK Biobank participants with dietary intake measured by 24-hour recall and genotyping. The primary predictors were dietary pattern, *PNPLA3*-rs738409-G, *TM6SF2*-rs58542926-T, a 16-variant hepatic steatosis polygenic risk score (PRS), and gene-environment interactions. The primary outcome was LFC, and secondary outcomes were iron-controlled T1 time (cT1, a measure of liver inflammation and fibrosis) and liver-related events/mortality.

Results: A total of 21,619 participants met inclusion criteria. In non-interaction models, Mediterranean diet and intake of fruit/vegetables/legumes and fish associated with lower LFC, while higher red/processed meat intake and all genetic predictors associated with higher LFC. In interaction models, all genetic predictors interacted with Mediterranean diet and fruit/vegetable/legume intake, while the steatosis PRS interacted with fish intake and the *TM6SF2* genotype interacted with red/processed meat intake, to affect LFC. Dietary effects on LFC were up to 3.8-fold higher in *PNPLA3*-rs738409-GG *vs.* –CC individuals, and 1.4-3.0-fold higher in the top *vs.* bottom quartile of the steatosis PRS. Gene-diet interactions were stronger in participants with *vs.* without overweight. The steatosis PRS interacted with Mediterranean diet and fruit/vegetable/legume intake to affect cT1 and most dietary and genetic predictors associated with risk of liver-related events or mortality by age 70.

Conclusions: Effects of diet on LFC and cT1 were markedly accentuated in patients at increased genetic risk for SLD, implying dietary interventions may be more impactful in these populations.

Introduction

Metabolic dysfunction-associated steatotic liver disease (MASLD) affects 20-40% of the population in developed regions across the world.[1–5] In the United States, MASLD is implicated in 7.5% of all deaths and 36% of liver-related deaths,[6] and is the fastest-growing cause of cirrhosis and associated complications including hepatocellular carcinoma.[7–9] There is a growing body of literature on associations between diet and MASLD risk. The rising prevalence of MASLD in many parts of the world has been correlated to westernization of dietary patterns. MASLD is associated with increased intake of diets high in carbohydrates (especially added sugars and sugar-sweetened beverages)[10,11] and animal proteins.[12,13] In contrast, a Mediterranean-style diet which is high in unsaturated fats but lower in saturated fats, carbohydrates, and red meat has been linked to improvement in hepatic steatosis.[14,15] The effect of other major components of diet such as fruits, vegetables, legumes, and fish is less well-established.

Heritability of hepatic steatosis has been estimated at 27-39%.[16,17] We and others have identified specific genetic variants associated with increased risk of MASLD and MASLD-related complications including cirrhosis and hepatocellular carcinoma, the best-established of which are *PNPLA3*-rs738409-G and *TM6SF2*-rs58542926-T.[17–24] In addition, polygenic risk scores (PRSs), which combine multiple genetic risk variants into a single continuous score, may capture the overall genetic risk for liver enzyme elevations, MASLD, or cirrhosis.[23,25] Recently, there has been increasing recognition that genetic risk interacts with environmental risk factors, and that individuals at higher genetic risk may derive increased benefit from interventions such as lifestyle modification.[26] We recently showed that genetic risk for MASLD interacts with insulin resistance and that the interaction between insulin

Keywords: SLD; gene-environment interaction; PNPLA3; TM6SF2; polygenic risk score; 24 hour recall; Mediterranean diet.
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concentration and a common genetic variant in *PNPLA3* associated with MASLD accounted for 8% of the variation in hepatic steatosis.[27] One recent study found that a PRS for hepatic steatosis predicted interval worsening of hepatic steatosis among individuals whose dietary quality worsened over time.[28] However, interactions between individual components of diet and individual genetic variants are less well-established.

In this study, we aimed to systematically evaluate the effects on hepatic steatosis of dietary factors and their interactions with genetic risk factors, specifically *PNPLA3*-rs738409-G, *TM6SF2*-rs58542926-T, and PRSs for hepatic steatosis, in a large, population-based cohort: the UK Biobank (UKBB).

Patients and methods

Cohorts, genotyping, and imputation

UKBB includes genotypic, clinical, and demographic information on over 500,000 individuals living in the United Kingdom aged 40-69 at the time of recruitment from 2006-2010. Genotyping and data collection has previously been described.[29] Participants were genotyped on one of two custom arrays: UK BiLEVE Axiom Array or UKBB Axiom Array with >95% overlap. Imputation was performed to UK10K and Haplotype Reference Consortium reference panels. As including closely related individuals can lead to overestimation of gene-environment interactions (type I error), we excluded closely related participants (second-degree or closer) using KING (Kinship-based Inference for Genome-wide association studies) version 2.1.5.[30] The primary MRI cohort included unrelated individuals with non-missing genotyping data, 24-hour recall surveys (per "Dietary variables" below), or other covariates ("Regression models and gene-environment interactions: Primary analysis" below). We did not exclude participants with significant alcohol intake because we did not observe heterogeneity of effects of the predictors across alcohol use categories (Tables S2, S6, and S7). We did however adjust for alcohol intake so that the results are more indicative of MASLD than alcohol-related liver disease.

Outcomes

Selected UKBB participants have undergone liver MRI, including quantification of liver fat content (LFC; primary outcome) with protein density fat fraction measurement[31] and iron-corrected T1 time (cT1; secondary outcome), a composite measure of liver inflammation and fibrosis which also predicts risk of liver-related complications.[32–34] We used convolutional neural networks to impute LFC for participants for whom liver fat images were available but reported LFC was not available, with excellent concordance (r^2 >0.95).[35]

Other secondary outcomes were liver-related events (LREs) and liver-related mortality. LREs were defined as an inpatient ICD-10 code of any of the following: K70.3, K74.5, K74.6, I85.0, I85.9, I86.44, K72.9, K65.2, K76.6, K76.7, or K76.81.[36] Liver-related mortality was defined as a code associated with cause of death of K7X. For both outcomes, cases were individuals experiencing the event before age 70 and controls were those who reached at least 70 years of age without experiencing the event, similar to the approach taken by a previous study.[25] We conducted logistic regression with cases and controls defined as above, and we evaluated the impact of dietary and genetic risk factors, either with no covariates (univariable analysis) or with the same covariates as in the primary analysis' model 1 (multivariable analysis). We did not conduct time-to-event analyses because the dates that 24-hour recall surveys were administered (the most plausible index date) were not available.

Dietary variables

Dietary patterns were determined based on the average of five 24-hour recall surveys; these recall questionnaires had a high level of concordance with food frequency questionnaires in UKBB and validated well compared to biomarkers.[37,38] We evaluated effects on hepatic steatosis of individual food types (specifically, fruit/vegetables/legumes, fish, and red/processed meat) and the modified Mediterranean diet score (MMED), which measures adherence to the Mediterranean diet in non-Mediterranean countries and correlates with cardiovascular disease and overall mortality.[39,40]

We analyzed MMED either as a continuous variable or based on quartiles. We measured individual food intake based on pre-specified cut-offs. Cut-offs for high *vs.* low intake of individual food types were determined based on the government-sponsored Public Health England Eatwell Guide (https://www.gov.uk/government/publications/the-eatwell-guide). The recommendations are as below: fruit, vegetables, and legumes: ≥5 servings/day; red or processed meat: <70 g/day; and fish: ≥2 servings/week with ≥1 serving/week of oily fish.

Regression models and gene-environment interactions: Primary analysis

We conducted linear regressions with LFC as the dependent variable. For univariable non-interaction models, the primary predictors were (1) dietary intake, defined as high *vs.* low intake of a specific food type, or each point of MMED, or (2) or genetic variables, namely dosage of *PNPLA3*-rs738409-G or *TM6SF2*-rs58542926-T, or a previously-reported PRS for MASLD.[35] The primary model was adjusted for age, sex, indices of multiple deprivation (a measure of poverty incorporating income, employment, education, healthcare, crime, housing, and living environment[41]), physical activity, number of alcohol-containing drinks per week, genetic principal components 1-10 to account for genetic ancestry, tobacco use (current, former, or never smoker), and total daily caloric intake. We also conducted numerous sensitivity analyses.

In stratified models, heterogeneity between the two groups was measured based on the Cochran Q statistic,[42] with p value for heterogeneity (P_{het}) <0.05 considered statistically significant.

Gene-environment (diet) interactions were determined in two ways. The first was using multiplicative interaction terms. For these, we constructed linear regression models with LFC as the dependent variable and covariates as above, and with the dependent variables being dietary intake, a genetic predictor, and the product of the genetic and dietary variables (*i.e.* $\beta_{diet} * diet + \beta_{gene} * gene + \beta_{d-g} * diet * gene$). The same models were generated as with the univariable analyses as described above. A significant gene-environment interaction was defined as p <0.05 for the interaction term. Heterogeneity in stratified gene-environment interaction models was measured as above. All interaction models utilized robust standard errors (HC0) to account for the heteroscedasticity expected in interaction models.[43]

The second way to identify gene-diet interactions was by evaluating heterogeneity in dietary effects across genetic strata.[44] The genetic strata used were (1) *PNPLA3*-rs738409-CC, -CG, *vs.* GG; (2) *TM6SF2*-rs58542926-CC *vs.* -CT/-TT; and (3) hepatic steatosis PRS quartile 1, 2 or 3, *vs.* 4 (Table S1).[35] As above, heterogeneity between the two groups was measured based on the Cochran Q statistic.[42]

Descriptive statistics

Quantitative variables were reported as mean ± standard error or median (interquartile range). Categorical variables were reported as percentages (%). All analyses were conducted in R version 3.5.1 (Vienna, Austria).

For non-interaction models and the primary outcome, we required a Bonferroni-corrected two-sided *p* value cut-off of <0.05 for statistical significance. For interaction models or secondary outcomes, we only included genetic and dietary predictors which were shown to be significant in univariable analyses, so we did not adjust for multiple testing, as these were established pre-specific hypotheses, and we used an unadjusted two-sided *p* value <0.05 for significance.

Ethics

All research in this study was approved by the Institutional Review Board of the University of Michigan (Ann Arbor, MI). UKBB protocols were approved by the National Research Ethics Service Committee and all participants provided written informed consent. Analyses in this project were conducted under UKBB Resource Project 18120.

Results

Descriptive statistics

The primary MRI cohort included 21,619 participants with non-missing data, of whom 3,841 (18%) had hepatic steatosis (LFC >5%) (Fig. 1). Those with hepatic steatosis were more frequently male and had higher BMI, transaminase, and triglyceride levels; lower high-density lipoprotein levels; and higher prevalence of diabetes and hypertension (Table 1). As expected, individuals with hepatic steatosis had higher dosages of *PNPLA3*-rs738409-G and *TM6SF2*-rs58542926-T alleles as well as higher steatosis PRS scores. In addition, hepatic steatosis was associated with greater intake of red/processed meat; lower intake of vegetables, fruit, and legumes; lower fish intake; and lower adherence to a Mediterranean diet pattern.

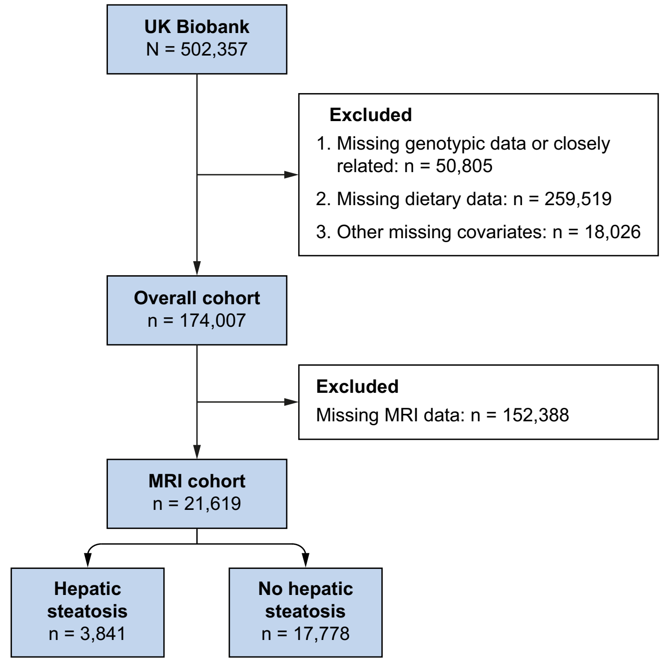

Fig. 1. Study inclusion and exclusion criteria. Hepatic steatosis is defined as liver fat content ≥5% on MRI proton density fat fraction. MRI, magnetic resonance imaging.

Univariable diet and genetic effects

In univariable analysis, we identified numerous associations between LFC and diet or genotypes (Table 2). Greater adherence to a Mediterranean-style diet associated with lower LFC (effect per quartile -0.30% [-0.35 to -0.25%]) while greater intake of red/processed meat associated with higher LFC (effect 0.47% [0.33-0.60%]). Higher intake of fruit, vegetables, or legumes associated with lower LFC (effect size -0.42% [-0.53 to -0.31%], as did higher fish intake (effect size -0.44% [-0.57 to -0.31%]). Consistent with previous findings, *PNPLA3*-rs738409-G and *TM6SF2*-rs58542926-T were both associated with higher LFC (per allele effect of 0.70% [0.59 to 0.81%] and 1.21% [1.01 to 1.41%], respectively). Similarly, higher steatosis PRS associated with increased LFC (effect per SD of steatosis PRS 0.53% [0.47-0.60%]). Note that the effect size for the PRS and individual variants are per SD and per allele, respectively, and are not directly comparable. Notably, a steatosis PRS excluding the *PNPLA3* and *TM6SF2* alleles remained associated with LFC (effect per SD of PRS 0.65% [0.59-0.71%], $p = 1.54 \times 10^{-99}$) suggesting a polygenic effect.

We conducted several sensitivity and subgroup analyses. In participants with *vs.* without significant alcohol use (>14 drinks/week for women or >21 for men[45]), all dietary and genetic variables remained significantly associated with LFC with no statistically significant heterogeneity between alcohol intake categories (P_{het} >0.05 for all) (Table S2). The primary analysis was not adjusted for BMI or diabetes, which are likely mediators of the effects of diet on hepatic steatosis. Adjusting for BMI and diabetes modestly reduced the effect size for all dietary predictors but did not affect effect size of genetic predictors (Table S3), consistent with the idea that BMI and diabetes mediate dietary but not genetic effects on hepatic steatosis. However, overweight participants (BMI ≥25 kg/m^2) had significantly larger effect sizes associated with MMED, red/processed meat, and fish intake than those of normal weight (BMI <25 kg/m^2) (Table S4). The difference in effect sizes in overweight *vs.* normal weight individuals were also striking for genetic predictors of *PNPLA3*-rs738409-G, *TM6SF2*-rs58542926-T, and steatosis PRS, consistent with our previous findings (Table S4).[27] When stratifying by sex, dietary effects were not significantly different in women *vs.* men (P_{het}>0.05 for all), but genetic effects were significantly larger in men (Table S5).

Table 1. Participant characteristics based on hepatic steatosis status.

Trait	Hepatic steatosis (n = 3,841)	No hepatic steatosis (n = 17,778)	*p* value
Demographics and comorbidities			
Age (years)	55.0 (0.1)	55.2 (0.1)	0.27
Male	60.9%	44.2%	<0.0001
Body mass index (kg/m^2)	29.6 (0.1)	25.7 (<0.1)	<0.0001
Alcoholic drinks/day	2.1 (<0.1)	1.8 (<0.1)	<0.0001
Significant alcohol intake	29.5%	29.4%	0.91
Tobacco use			
Never	43.2%	44.6%	0.12
Former	54.2%	53.2%	0.30
Current	2.7%	2.2%	0.12
Diabetes mellitus	14.1%	4.5%	<0.0001
Hypertension	60.6%	41.8%	<0.0001
Laboratory values			
Total cholesterol (mmol/L)	5.7 (<0.1)	5.7 (<0.1)	0.27
Low-density lipoprotein (mmol/L)	3.7 (<0.1)	3.6 (<0.1)	<0.0001
High-density lipoprotein (mmol/L)	1.3 (<0.1)	1.5 (<0.1)	<0.0001
Triglycerides (mmol/L)	2.1 (<0.1)	1.5 (<0.1)	<0.0001
Hemoglobin A1c (mmol/mol)	36.5 (0.1)	34.7 (<0.1)	<0.0001
ALT (U/L)	30.7 (0.3)	21.1 (0.9)	<0.0001
AST (U/L)	28.5 (0.2)	25.2 (0.1)	<0.0001
Albumin (g/L)	45.7 (<0.1)	45.5 (<0.1)	<0.0001
Dietary patterns			
Modified Mediterranean diet score			
0-1	14.7%	9.8%	<0.0001
2	22.1%	18.3%	
3	25.8%	24.1%	
4	21.4%	23.5%	
5	11.5%	16.1%	
6	3.9%	6.8%	
7-8	0.8%	1.4%	
High fruit, vegetable, and legume intake	37.6%	45.9%	<0.0001
High red/processed meat intake	34.6%	27.3%	<0.0001
High fish intake	18.2%	23.1%	<0.0001
Genetic risk factors			
Steatosis polygenic risk score (rank units)	0.24 (0.02)	-0.04 (0.01)	<0.0001
PNPLA3-rs738409 genotype			
CC	52.9%	63.5%	<0.0001
CG	39.6%	32.5%	
GG	7.4%	4.0%	
TM6SF2-rs58542926 genotype			
CC	79.9%	87.0%	<0.0001
CT	18.7%	12.6%	
TT	1.4%	0.4%	

Data are reported as mean (standard error) or percentage (%). Continuous values were compared using a Student's *t* test and categorical variables using a chi-square test.

Gene-environment interactions

Next, we generated interaction models to evaluate whether there were interactions between genetic and dietary predictors, *i.e.* whether dietary effects are different based on genetic risk (Methods). Using multiplicative interaction models, we identified significant interactions between the steatosis PRS and MMED, red/processed meat intake, fruit/vegetable/legume intake, and fish intake (Table 3; p <0.05 for all). We also found

Table 2. Univariable effects of genetic and dietary predictors.

	All participants (N = 21,619)	
Predictor	**Effect**	***p* value**
Modified Mediterranean diet score (per quartile)	-0.30 (-0.35 to -0.25)	3.66E-29
Vegetable, fruit, and legume intake (high *vs.* low)	-0.42 (-0.53 to -0.31)	4.95E-12
Red/processed meat intake (high *vs.* low)	0.47 (0.33 to 0.60)	3.01E-11
Fish intake (high *vs.* low)	-0.44 (-0.57 to -0.31)	4.11E-10
Steatosis polygenic risk score (per standard deviation)	0.53 (0.47 to 0.60)	2.65E-63
PNPLA3-rs738409 (per G allele)	0.70 (0.59 to 0.81)	8.70E-37
TM6SF2-rs58542926 (per T allele)	1.21 (1.01 to 1.41)	3.23E-32

Models were constructed as linear regressions with liver fat content as the dependent variable, and the predictor as the primary independent variable. All models were adjusted for age, sex, genetic principal components 1-10 (to account for genetic ancestry), physical activity index, Indices of Multiple Deprivation, number of alcoholic drinks/week, tobacco status (active *vs.* former *vs.* never smoker), and total caloric intake. Effects are reported as effect on liver fat content (95% confidence interval). Positive effect implies that predictor is associated with increased liver fat content. *P* values were Bonferroni-adjusted for seven comparisons.

that the PRS, *PNPLA3*-rs738409-G, and *TM6SF2*-rs58542926-T all interacted with Mediterranean diet adherence and vegetable/fruit/legume intake (p <0.05 for all) to amplify dietary effects (Table 3, Fig. 2). The PRS and *TM6SF2* genotype also interacted with fish intake and red/processed meat intake, respectively (Table 3).

When we generated heterogeneity models to identify gene-environment interactions, the interactions were concordant with the multiplicative models (Table 4), with the exception of interactions between the PRS and vegetable/fruit/legume intake which was now only borderline significant (p = 0.058). The effects of dietary predictors on risk of MASLD (defined as LFC >5%) were also significantly different across different genetic risk categories (Fig. 3).

The magnitude of these gene-diet interactions was large. Effects of diet on LFC in steatosis PRS quartile 4 were 1.4- to 3.0-fold higher than the effects in quartile 1 (*e.g.* effect of high *vs.* low fish intake was -0.71 *vs.* -0.24 in quartile 4 *vs.* 1; Table 4). Likewise, in *PNPLA3*-rs738409-GG *vs.* –CC carriers, the effect of high vegetable/fruit/legume intake was 3.8-fold higher (effect size -1.12 *vs.* -0.35%), and the per-quartile effect of MMED was -0.48 *vs.* -0.25% (1.9-fold; Table 4).

These patterns remained when we evaluated the continuous trait of prevalence of hepatic steatosis, defined as LFC ≥5%. Among participants in the top quartile of steatosis PRS, 18.6% with MMED in the top quartile had LFC >5% compared to 30.3% with MMED in the bottom quartile (difference = 11.7%); in those with steatosis PRS in the bottom percentile, the corresponding values are 18.4% and 10.5% (difference = 7.9%; Fig. 2). Among participants with *PNPLA3*-rs738409-CC genotype and MMED score in the bottom quartile, 20.0% had LFC >5% compared to 11.5% with MMED in the top quartile (difference = 8.5%); in *PNPLA3*-rs738409-GG participants, the values are 35.4% and 21.7% (difference = 13.7%; Fig. 2).

We conducted a number of sensitivity analyses with the multiplicative gene-environment interaction models. First, we separately evaluated participants without (Table S6) and with (Table S7) significant alcohol intake. As before with non-interaction models (Table S2), there was no statistically significant heterogeneity based on alcohol intake. Second, we adjusted for BMI and diabetes mellitus (Table S8). When this was done, most gene-environment interactions decreased modestly in statistical significance, as expected. Third, we stratified by BMI category, BMI <25 kg/m^2 *vs.* ≥25 kg/m^2. Notably, while nearly all of the significant gene-environment interactions from the primary analysis remained statistically significant in normal weight individuals (Table S9), only *TM6SF2* genotype and red/processed meat effects interacted in overweight participants (Table S10). Gene main term effects were universally stronger in overweight participants, as were some main dietary effects (Table S10). We evaluated sex-specific interactions in women (Table S12) or men (Table S12), and we found heterogeneity in a few PRS and *TM6SF2* main genetic terms, consistent with our previous findings. Finally, we evaluated the effects of a modified steatosis PRS, which excluded the *PNPLA3* and *TM6SF2* alleles (Table S12). The modified PRS interacted with vegetable/fruit/legume and red/processed meat intake, and had a borderline significant interaction with MMED (p = 0.070), suggesting that the gene-environment interactions in the steatosis PRS were not solely driven by just the *PNPLA3* and *TM6SF2* alleles and that there is polygenicity in gene-diet interactions.

Secondary outcomes: iron-corrected T1 time, liver-related events, and liver-related mortality

Next, we evaluated whether dietary and genetic predictors influenced cT1, a combined measure of fibrosis and inflammation.[33] In univariable analyses, all of the dietary and genetic predictors were associated with cT1 in the same direction that they were associated with LFC (Table S4). In interaction models including genetic main, dietary main, and gene-diet interaction

Table 3. Effects of interactions between genetic and dietary predictors on hepatic steatosis.

	Genetic main term		Dietary main term		Gene-diet interaction term	
Dietary variable	**Effect**	***p* value**	**Effect**	***p* value**	**Effect**	***p* value**
Steatosis polygenic risk score (per standard deviation) (n = 21,619)						
Modified mediterranean diet score (per quartile) *	0.72 (0.55 to 0.88)	<0.0001	-0.30 (-0.35 to -0.25)	<0.0001	-0.07 (-0.13 to -0.02)	**0.012**
Vegetable, fruit, and legume intake (high *vs.* low) *	0.45 (0.37 to 0.53)	<0.0001	-0.41 (-0.53 to -0.30)	<0.0001	-0.15 (-0.27 to -0.03)	**0.014**
Red/processed meat intake (high *vs.* low)	0.50 (0.43 to 0.57)	<0.0001	0.46 (0.33 to 0.59)	<0.0001	0.11 (-0.04 to 0.25)	0.14
Fish intake (high *vs.* low) *	0.41 (0.29 to 0.53)	<0.0001	-0.43 (-0.56 to -0.30)	<0.0001	-0.18 (-0.32 to -0.04)	**0.013**
***PNPLA3*-rs738409 (per G allele) (n=21,619)**						
Modified mediterranean diet score (per quartile) *	0.98 (0.69 to 1.27)	<0.0001	-0.25 (-0.31 to -0.19)	<0.0001	-0.11 (-0.21 to -0.01)	**0.029**
Vegetable, fruit, and legume intake (high *vs.* low) *	0.56 (0.42 to 0.71)	<0.0001	-0.32 (-0.45 to -0.19)	<0.0001	-0.26 (-0.47 to -0.04)	**0.018**
Red/processed meat intake (high *vs.* low)	0.64 (0.52 to 0.76)	<0.0001	0.38 (0.23 to 0.53)	<0.0001	0.21 (-0.04 to 0.45)	0.11
Fish intake (high *vs.* low)	0.69 (0.47 to 0.92)	<0.0001	-0.44 (-0.59 to -0.29)	<0.0001	0.01 (-0.25 to 0.27)	0.93
***TM6SF2*-rs58542926 (per T allele) (n=21,619)**						
Modified mediterranean diet score (per quartile) *	1.80 (1.27 to 2.33)	<0.0001	-0.27 (-0.32 to -0.22)	<0.0001	-0.23 (-0.41 to -0.05)	**0.013**
Vegetable, fruit, and legume intake (high *vs.* low) *	0.93 (0.66 to 1.21)	<0.0001	-0.34 (-0.46 to -0.23)	<0.0001	-0.48 (-0.87 to -0.09)	**0.016**
Red/processed meat intake (high *vs.* low) *	1.06 (0.84 to 1.29)	<0.0001	0.40 (0.26 to 0.53)	<0.0001	0.53 (0.05 to 1.00)	**0.029**
Fish intake (high *vs.* low)	1.01 (0.60 to 1.41)	<0.0001	-0.40 (-0.53 to -0.27)	<0.0001	-0.27 (-0.73 to 0.20)	0.26

Models were constructed as linear regressions with liver fat content as the dependent variable, and three primary predictors: (1) genetic main term, (2) dietary main term, and (3) gene-diet interaction term defined as genetic*dietary term. All models were adjusted for age, sex, genetic principal components 1-10 (to account for genetic ancestry), physical activity index, Indices of Multiple Deprivation, number of alcoholic drinks/week, tobacco status (active *vs.* former *vs.* never smoker), and total caloric intake. Effects are reported as effect on liver fat content (95% CI). Positive effect implies that predictor is associated with increased liver fat content. *P* values were by linear regression. * indicates that there is a significant gene-diet interaction (p <0.05). Bold text indicates a significant P_{het} value.

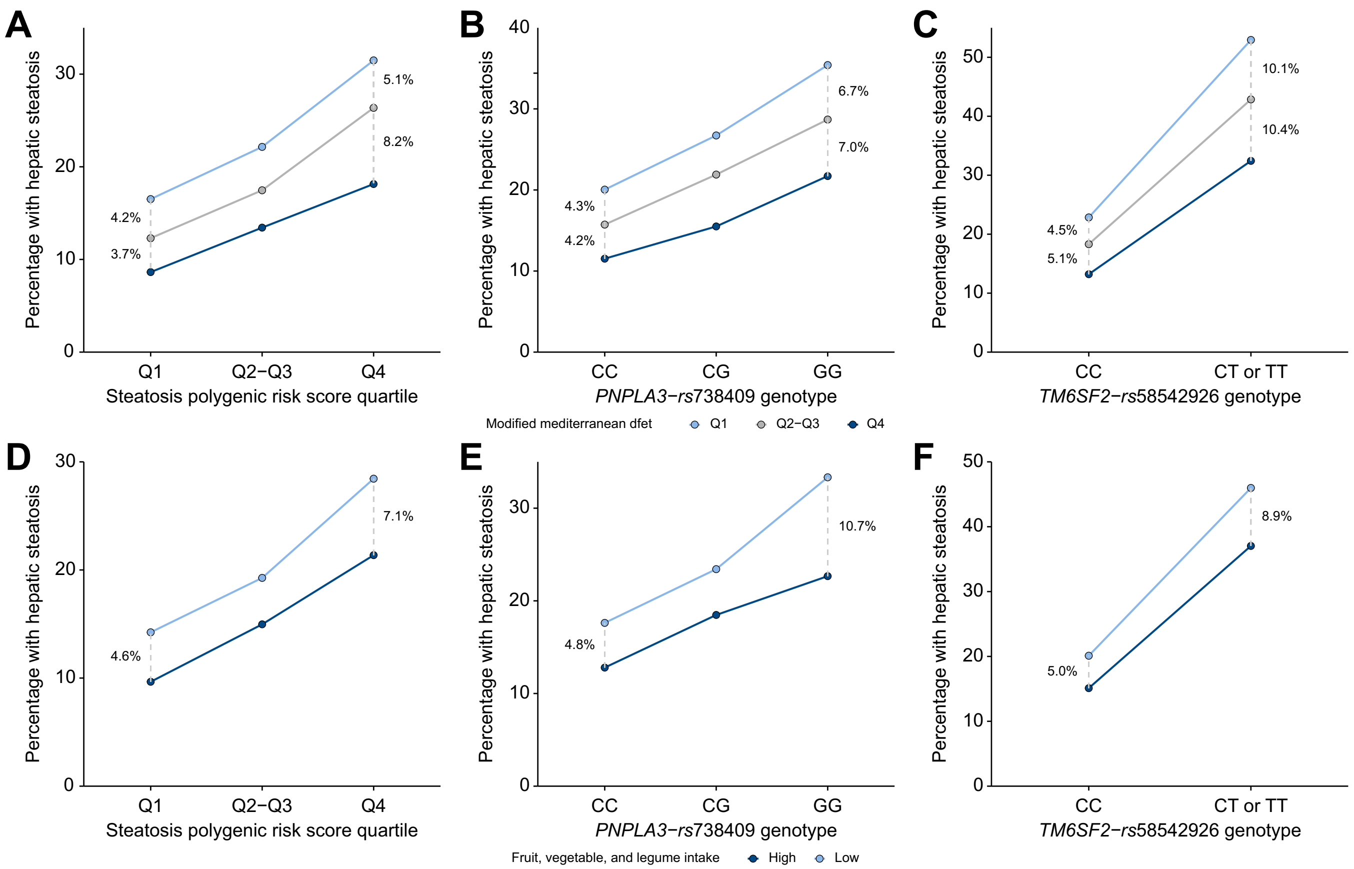

Fig. 2. Prevalence of hepatic steatosis based on dietary patterns and genotype. Line color represents dietary pattern, defined as (A-C) modified Mediterranean diet score quartile or (D-F) intake of fruit, vegetables, and legumes with cut-offs defined as ≥ or <5 servings/day per Public Health England Eatwell criteria. The x-axis represents genetic risk, specifically (A, D) steatosis polygenic risk score quartile (Q), (B, E) *PNPLA3*-rs738409 genotype, or (C, F) *TM6SF2*-rs58542926 genotype. The y-axis represents the percentage of participants with hepatic steatosis, defined as liver fat content ≥5%.

terms, the PRS associated with MMED and vegetable/fruit/legume intake to multiplicatively decrease cT1 times (Table S15). *PNPLA3* and *TM6SF2* genotypes did not interact with dietary predictors (Table S15).

Finally, we assessed associations between dietary and genetic predictors and risk of LREs or liver-related mortality by age 70 (Fig. 3; Table S16). In univariable analysis, higher MMED score and higher vegetable/fruit/legume intake were associated with lower risk of both LREs and liver-related mortality, while the steatosis PRS and *TM6SF2* genotype were associated with a higher risk of both outcomes (Table S16). After adjustment for covariates, the associations between dietary/genetic predictors and LREs were largely attenuated but some associations remained significant: MMED associated with lower risk of LREs and higher steatosis PRS and *TM6SF2*-rs58542926-T dosage with higher risk; the only predictor still associated with increased liver-related mortality was *PNPLA3*-rs738409-G dosage (Table S16). There were no significant gene-diet interactions for either LREs or liver-related mortality (data not shown).

Discussion

We comprehensively evaluated the impact of gene-diet interactions and their effects on LFC in a large, population-based cohort. We identified both dietary patterns (Mediterranean diet) and individual components of diet (red/processed meat, vegetables, fruit) that associated with LFC and whose effects were accentuated by genetic risk. Effects of diet on LFC differed greatly depending on genetic risk and were up to 3.8-fold higher in those with *PNPLA3*-rs738409-GG *vs.* –CC genotypes and 1.4-3.0-fold higher in the top *vs.* bottom quartile of the steatosis PRS. Additionally, interactions between the steatosis PRS and dietary factors influenced cT1 time, and both dietary and genetic predictors were associated with LREs and liver-related mortality.

Our findings are largely concordant with other studies evaluating diet in MASLD. Multiple studies have shown that specific eating patterns as measured by Mediterranean diet scales and the alternate healthy eating index are linked to decreased hepatic steatosis.[15,28] Certain dietary components such as high intake of red and processed meat, low intake of dietary fiber, and sugar-sweetened beverages are linked to increased hepatic steatosis.[11,12,46,47] Recent AASLD guidelines on the management of MASLD recommend weight loss but note that there is insufficient data to recommend any specific type of diet.[45] In contrast to the robust literature on dietary patterns and risk for MASLD, the literature on gene-diet interactions is comparatively limited. One study found that changes in diet over time (with diet quality measured by alternate healthy eating index and Mediterranean diet score) associated with changes in hepatic steatosis over time, and that this

Table 4. Heterogeneity in effects of diet on hepatic steatosis based on genetic risk.

Steatosis polygenic risk score	**Quartile 1, n = 5,351**		**Quartiles 2-3, n = 10,827**		**Quartile 4, n = 5,441**		
Dietary variable	**Dietary effect**	***p* value**	**Dietary effect**	***p* value**	**Dietary effect**	***p* value**	P_{het}
Modified mediterranean diet score (per quartile) *	-0.25 (-0.34 to -0.16)	<0.0001	-0.26 (-0.33 to -0.19)	<0.0001	-0.43 (-0.56 to -0.30)	<0.0001	**0.014**
Vegetable, fruit, and legume intake (high *vs.* low)	-0.35 (-0.53 to -0.15)	<0.0001	-0.29 (-0.45 to -0.13)	0.00027	-0.60 (-0.88 to -0.32)	<0.0001	0.058
Red/processed meat intake (high *vs.* low)	0.36 (0.14 to 0.58)	0.0015	0.36 (0.18 to 0.53)	<0.0001	0.49 (0.17 to 0.80)	0.0024	0.46
Fish intake (high *vs.* low) *	-0.24 (-0.47 to -0.02)	0.035	-0.47 (-0.64 to -0.30)	<0.0001	-0.71 (-1.04 to -0.39)	<0.0001	**0.016**

***PNPLA3*-rs738409 genotype**	**CC, n = 13,324**		**CG, n = 7,292**		**GG, n = 1,003**		P_{het}
Dietary variable	**Dietary effect**	***p* value**	**Dietary effect**	***p* value**	**Dietary effect**	***p* value**	
Modified mediterranean diet score (per quartile) *	-0.25 (-0.31 to -0.19)	<0.0001	-0.36 (-0.46 to -0.27)	<0.0001	-0.48 (-0.81 to -0.14)	0.0058	**0.031**
Vegetable, fruit, and legume intake (high *vs.* low) *	-0.35 (-0.48 to -0.22)	<0.0001	-0.47 (-0.68 to -0.26)	<0.0001	-1.12 (-1.81 to -0.43)	0.0015	**0.025**
Red/processed meat intake (high *vs.* low)	0.38 (0.23 to 0.53)	<0.0001	0.61 (0.36 to 0.86)	<0.0001	0.56 (-0.24 to 1.36)	0.17	0.11
Fish intake (high *vs.* low)	-0.45 (-0.60 to -0.30)	<0.0001	-0.40 (-0.65 to -0.15)	0.0019	-0.66 (-1.49 to 0.17)	0.12	0.52

***TM6SF2*-rs58542926 genotype**	**CC, n = 18,538**		**CT or TT, n = 3,081**		P_{het}
Dietary variable	**Dietary effect**	***p* value**	**Dietary effect**	***p* value**	
Modified mediterranean diet score (per quartile) *	-0.27 (-0.33 to -0.22)	<0.0001	-0.48 (-0.66 to -0.31)	<0.0001	**0.025**
Vegetable, fruit, and legume intake (high *vs.* low) *	-0.35 (-0.47 to -0.24)	<0.0001	-0.78 (-1.17 to -0.39)	<0.0001	**0.038**
Red/processed meat intake (high *vs.* low) *	0.40 (0.27 to 0.53)	<0.0001	0.97 (0.51 to 1.42)	<0.0001	**0.020**
Fish intake (high *vs.* low)	-0.40 (-0.53 to -0.26)	<0.0001	-0.68 (-1.12 to -0.24)	0.0026	0.23

Models were constructed as linear regressions with liver fat content as the dependent variable and dietary patterns as the primary predictor, stratified by genetic background. All models were adjusted for age, sex, genetic principal components 1-10 (to account for genetic ancestry), physical activity index, Indices of Multiple Deprivation, number of alcoholic drinks/week, tobacco status (active *vs.* former *vs.* never smoker), and total caloric intake. Effects are reported as effect on liver fat content (95% CI). Positive effect implies that predictor is associated with increased liver fat content. *P* values were by linear regression. *P* value for heterogeneity (P_{het}) was defined by the Cochran Q statistic; P_{het} <0.05 was used as the cut-off for a significant gene-diet interaction and is indicated by *. Bold text indicates a significant P_{het} value.

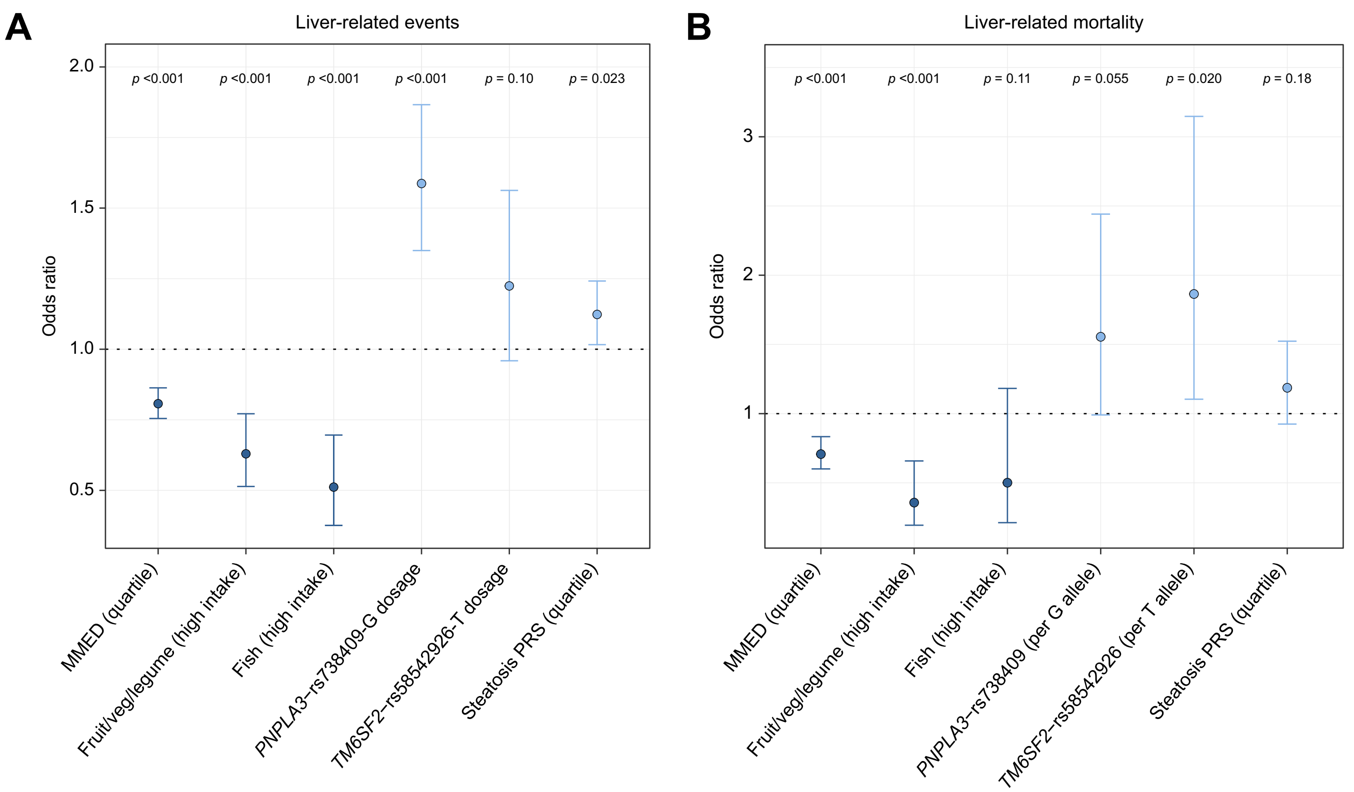

Fig. 3. Effects of dietary and genetic predictors on risk of liver-related events or liver-related mortality. (A) Effects of dietary and genetic predictors on risk of developing liver-related events by age 70. Dots represent the odds ratio and error bars represent the 95% CI based on a univariable logistic regression model. The predictors are the MMED score (effect is shown per quartile of MMED score), high intake of fruits, vegetables, and legumes (fruit/veg/legume), high intake of fish, *PNPLA3*-rs738409 genotype (effect per G allele), *TM6SF2*-rs58542926 genotype (effect per T allele), and the steatosis PRS (effect per quartile). Levels of significance: p <0.001 for MMED, fruit/vegetable/legume intake, fish intake, and *PNPLA3* genotype; p = 0.10 for *TM6SF2* genotype; p = 0.023 for steatosis PRS. (B) Effects of dietary and genetic predictors on risk of liver-related mortality by age 70. Levels of significance: p <0.001 for MMED and fruit/vegetable/legume intake; p = 0.11 for fish intake; p = 0.055 for *PNPLA3* genotype; p = 0.20 for *TM6SF2* genotype; p = 0.18 for steatosis PRS. p values were determined by logistic regression. MMED, modified Mediterranean diet score; PRS, polygenic risk score.

effect was primarily seen in persons at increased genetic risk (defined by a PRS).[28] The longitudinal nature of this study suggests that gene-environment interactions may influence steatosis progression over time. Other studies have evaluated interactions between *PNPLA3* rs738409-G and intake of refined carbohydrates, fruits, or polyunsaturated fat and their effects on hepatic steatosis and, to a lesser extent, fibrosis.[48–52] Due to its larger size and detailed clinical data, we implicated additional dietary groups including red/processed meat and vegetable intake. In addition, there are few published studies on interactions between *TM6SF2* variants and diet,[53] and this study is to our knowledge the first to report interactions between these variants and Mediterranean diet, fruit, or vegetable intake.

While dietary optimization is recommended for treatment of hepatic steatosis, it is typically intensive lifestyle intervention that is most effective in leading to substantial, sustained improvements in diet and weight.[54,55] However, such intensive interventions are not universally available and it is unlikely that this could be scaled to the population level given the high prevalence of MASLD. Our findings suggest that the greatest benefits in dietary modification may be seen in those at elevated baseline genetic risk. As direct-to-consumer genotyping becomes increasingly common, in the future it may be beneficial to prioritize patients at increased genetic risk (based on individual variants or PRSs) for intensive lifestyle interventions, as a step toward precision health.

Our analyses of effects of genetics and diet in groups stratified by BMI and alcohol intake yielded different findings with biological significance. The main genetic and dietary terms were mostly larger in overweight individuals, consistent with a "two-hit" hypothesis in which having multiple risk factors for hepatic steatosis may increase disease risk.[27] We note that there was only one significant gene-diet interaction among the overweight, though a number were present in the normal weight group. The reasons for this are not entirely clear but it may be that BMI partially mediates dietary effects on hepatic steatosis (as seen in difference in effect sizes with *vs.* without adjustment for BMI); that is, poor dietary quality increases risk of hepatic steatosis in part because it increases BMI. Of note, diet still influenced hepatic steatosis after adjusting for BMI, suggesting that improving diet quality may improve severity of hepatic steatosis even if weight loss is not achieved,[46] *especially* in those at higher genetic risk. In contrast, there was no evidence of heterogeneity in genetic effects, dietary effects, or gene-environment interactions across categories of alcohol intake. Thus, we did not find strong evidence that alcohol mediates dietary effects or gene-diet interactions.

Limitations of this study include that it was primarily a European-ancestry cohort based on the United Kingdom, which may limit reproducibility in other ancestries. Alcohol intake may be underreported. There were no significant gene-diet interactions on LREs or mortality, likely due to inadequate power (*i.e.* low incidence rate of LREs in UKBB); we might observe gene-diet interactions in cohorts with higher incidences of liver-related complications. Finally, steatosis was measured on MRI at a single time point while dietary patterns were measured and averaged at multiple times; however, others have reported that responses to dietary questionnaires in UKBB are stable over up to 4 years.[56] Strengths include that this was a large population-based cohort with a robust definition of hepatic steatosis based on proton density fat fraction.

In conclusion, we comprehensively evaluated the effects of diet and genetics, alcohol intake, and BMI on hepatic steatosis, cT1, and liver-related outcomes in a population-based cohort. Dietary patterns may influence severity of hepatic steatosis and inflammation/fibrosis even more in people at increased genetic risk than in those at low or average risk, and thus those with elevated genetic risk for hepatic steatosis or inflammation/fibrosis may benefit more from following dietary recommendations.

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Abbreviations

cT1, iron-corrected T1 time; LFC, liver fat content; MASLD, metabolic dysfunction-associated steatotic liver disease; MMED, modified Mediterranean diet score; MRI, magnetic resonance imaging; P_{het}, P value for heterogeneity; PRS, polygenic risk score; UKBB, UK Biobank.

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Conflict of interest

The authors declare no potential competing interests related to this study.

Please refer to the accompanying ICMJE disclosure forms for further details.

Authors' contributions

Vincent Chen conducted study design, data analysis and interpretation, and drafting of the manuscript. Xiaomeng Du, Yanhua Chen, Antonino Oliveri, Annapurna Kuppa, Brian Halligan, and Michael Province conducted data analysis and interpretation, and critical review of the manuscript. Elizabeth Speliotes conducted concept development, study design, data analysis and interpretation, writing of the manuscript and critical revision of the manuscript. All authors identified above have critically reviewed the paper and approve the final version of this paper, including the authorship statement. Guarantor of Article: Elizabeth Speliotes.

Data availability statement

Individual-level data will not be made publicly available due to UK Biobank privacy restrictions. UK Biobank data are available to qualified investigators upon application (https://www.ukbiobank.ac.uk/enable-your-research/apply-for-access).

Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jhep.2024.03.045.

References

Author names in bold designate shared co-first authorship

[1] Le MH, Devaki P, Ha NB, et al. Prevalence of non-alcoholic fatty liver disease and risk factors for advanced fibrosis and mortality in the United States. PLoS One 2017;12:e0173499.

[2] Lazo M, Hernaez R, Eberhardt MS, et al. Prevalence of nonalcoholic fatty liver disease in the United States: the third national health and nutrition examination survey, 1988-1994. Am J Epidemiol 2013;178:38–45.

[3] Williams CD, Stengel J, Asike MI, et al. Prevalence of nonalcoholic fatty liver disease and nonalcoholic steatohepatitis among a largely middle-aged population utilizing ultrasound and liver biopsy: a prospective study. Gastroenterology 2011;140:124–131.

[4] Eguchi Y, Hyogo H, Ono M, et al. Prevalence and associated metabolic factors of nonalcoholic fatty liver disease in the general population from 2009 to 2010 in Japan: a multicenter large retrospective study. J Gastroenterol 2012;47:586–595.

[5] Wong VW, Chu WC, Wong GL, et al. Prevalence of non-alcoholic fatty liver disease and advanced fibrosis in Hong Kong Chinese: a population study using proton-magnetic resonance spectroscopy and transient elastography. Gut 2012;61:409–415.

[6] Alvarez CS, Graubard BI, Thistle JE, et al. Attributable fractions of NAFLD for mortality in the United States: results from NHANES III with 27 Years of follow-up. Hepatology 2020;72:430–440.

[7] Kim D, Li AA, Perumpail BJ, et al. Changing trends in etiology-based and ethnicity-based annual mortality rates of cirrhosis and hepatocellular carcinoma in the United States. Hepatology 2019;69:1064–1074.

[8] Wong RJ, Aguilar M, Cheung R, et al. Nonalcoholic steatohepatitis is the second leading etiology of liver disease among adults awaiting liver transplantation in the United States. Gastroenterology 2015;148:547–555.

[9] Wong RJ, Cheung R, Ahmed A. Nonalcoholic steatohepatitis is the most rapidly growing indication for liver transplantation in patients with hepatocellular carcinoma in the U.S. Hepatology 2014;59:2188–2195.

[10] Ma J, Fox CS, Jacques PF, et al. Sugar-sweetened beverage, diet soda, and fatty liver disease in the Framingham Heart Study cohorts. J Hepatol 2015;63:462–469.

[11] Jia Q, Xia Y, Zhang Q, et al. Dietary patterns are associated with prevalence of fatty liver disease in adults. Eur J Clin Nutr 2015;69:914–921.

[12] Noureddin M, Zelber-Sagi S, Wilkens LR, et al. Diet associations with nonalcoholic fatty liver disease in an ethnically diverse population: the Multiethnic Cohort. Hepatology 2020;71:1940–1952.

[13] Yang CQ, Shu L, Wang S, et al. Dietary patterns modulate the risk of non-alcoholic fatty liver disease in Chinese adults. Nutrients 2015;7:4778–4791.

[14] Yaskolka Meir A, Rinott E, Tsaban G, et al. Effect of green-Mediterranean diet on intrahepatic fat: the DIRECT PLUS randomised controlled trial. Gut 2021;70:2085–2095.

[15] Ryan MC, Itsiopoulos C, Thodis T, et al. The Mediterranean diet improves hepatic steatosis and insulin sensitivity in individuals with non-alcoholic fatty liver disease. J Hepatol 2013;59:138–143.

[16] Schwimmer JB, Celedon MA, Lavine JE, et al. Heritability of nonalcoholic fatty liver disease. Gastroenterology 2009;136:1585–1592.

[17] **Speliotes EK**, **Yerges-Armstrong LM**, **Wu J**, et al. Genome-wide association analysis identifies variants associated with nonalcoholic fatty liver disease that have distinct effects on metabolic traits. Plos Genet 2011;7:e1001324.

[18] Gellert-Kristensen H, Richardson TG, Davey Smith G, et al. Combined effect of PNPLA3, TM6SF2, and HSD17B13 variants on risk of cirrhosis and hepatocellular carcinoma in the general population. Hepatology 2020;72:845–856.

[19] Chen VL, Wright AP, Halligan B, et al. Body composition and genetic lipodystrophy risk score associate with nonalcoholic fatty liver disease and liver fibrosis. Hepatol Commun 2019;3:1073–1084.

[20] Jamialahmadi O, Mancina RM, Ciociola E, et al. Exome-wide association study on alanine aminotransferase identifies sequence variants in the GPAM and APOE associated with fatty liver disease. Gastroenterology 2021;160:1634–1646.

[21] Chen VL, Chen Y, Du X, et al. Genetic variants that associate with cirrhosis have pleiotropic effects on human traits. Liver Int 2019;40:405–415.

[22] Trépo E, Nahon P, Bontempi G, et al. Association between the PNPLA3 (rs738409 C>G) variant and hepatocellular carcinoma: evidence from a

meta-analysis of individual participant data. Hepatology 2014;59: 2170–2177.
[23] **Chen VL**, **Du X**, Chen Y, et al. Genome-wide association study of serum liver enzymes implicates diverse metabolic and liver pathology. Nat Commun 2021;12:816.
[24] Chalasani N, Guo X, Loomba R, et al. Genome-wide association study identifies variants associated with histologic features of nonalcoholic fatty liver disease. Gastroenterology 2010;139:1567–1576. 1576 e1561-1566.
[25] Emdin CA, Haas M, Ajmera V, et al. Association of genetic variation with cirrhosis: a multi-trait genome-wide association and gene-environment interaction study. Gastroenterology 2020;160:1620–1633.
[26] Khera AV, Emdin CA, Drake I, et al. Genetic risk, adherence to a healthy lifestyle, and coronary disease. N Engl J Med 2016;375:2349–2358.
[27] Barata L, Feitosa MF, Bielak LF, et al. Insulin resistance exacerbates genetic predisposition to nonalcoholic fatty liver disease in individuals without diabetes. Hepatol Commun 2019;3:894–907.
[28] Ma J, Hennein R, Liu C, et al. Improved diet quality associates with reduction in liver fat, particularly in individuals with high genetic risk scores for nonalcoholic fatty liver disease. Gastroenterology 2018;155:107–117.
[29] Sudlow C, Gallacher J, Allen N, et al. UK biobank: an open access resource for identifying the causes of a wide range of complex diseases of middle and old age. Plos Med 2015;12:e1001779.
[30] Manichaikul A, Mychaleckyj JC, Rich SS, et al. Robust relationship inference in genome-wide association studies. Bioinformatics 2010;26:2867–2873.
[31] Wilman HR, Kelly M, Garratt S, et al. Characterisation of liver fat in the UK Biobank cohort. PLoS One 2017;12:e0172921.
[32] **Parisinos CA**, **Wilman HR**, Thomas EL, et al. Genome-wide and Mendelian randomisation studies of liver MRI yield insights into the pathogenesis of steatohepatitis. J Hepatol 2020;73:241–251.
[33] Andersson A, Kelly M, Imajo K, et al. Clinical utility of magnetic resonance imaging biomarkers for identifying nonalcoholic steatohepatitis patients at high risk of progression: a multicenter pooled data and meta-analysis. Clin Gastroenterol Hepatol 2021;20:2451–2461.
[34] Jayaswal ANA, Levick C, Selvaraj EA, et al. Prognostic value of multiparametric magnetic resonance imaging, transient elastography and blood-based fibrosis markers in patients with chronic liver disease. Liver Int 2020;40:3071–3082.
[35] Chen Y, Du X, Kuppa A, et al. Genome-wide association meta-analysis identifies 17 loci associated with nonalcoholic fatty liver disease. Nat Genet 2023;55:1640–1650.
[36] Chen VL, Oliveri A, Miller MJ, et al. PNPLA3 genotype and diabetes identify patients with non-alcoholic fatty liver disease at high risk of incident cirrhosis. Gastroenterology 2023;164:966–977.
[37] Greenwood DC, Hardie LJ, Frost GS, et al. Validation of the oxford WebQ online 24-hour dietary questionnaire using biomarkers. Am J Epidemiol 2019;188:1858–1867.
[38] Carter JL, Lewington S, Piernas C, et al. Reproducibility of dietary intakes of macronutrients, specific food groups, and dietary patterns in 211 050 adults in the UK Biobank study. J Nutr Sci 2019;8:e34.
[39] Tektonidis TG, Akesson A, Gigante B, et al. A Mediterranean diet and risk of myocardial infarction, heart failure and stroke: a population-based cohort study. Atherosclerosis 2015;243:93–98.
[40] Warensjo Lemming E, Byberg L, Wolk A, et al. A comparison between two healthy diet scores, the modified Mediterranean diet score and the Healthy Nordic Food Index, in relation to all-cause and cause-specific mortality. Br J Nutr 2018;119:836–846.
[41] Tyrrell J, Jones SE, Beaumont R, et al. Height, body mass index, and socioeconomic status: mendelian randomisation study in UK Biobank. BMJ 2016;352:i582.
[42] Cochran WG. The combination of estimates from different experiments. Biometrics 1954;10:101–129.
[43] White H. A heteroskedasticity-consistent covariance matrix estimator and a direct test for heteroskedasticity. Econometrica 1980;48:817–838.
[44] Kraft P, Yen Y-C, Stram DO, et al. Exploiting gene-environment interaction to detect genetic associations. Hum Hered 2007;63:111–119.
[45] Chalasani N, Younossi Z, Lavine JE, et al. The diagnosis and management of nonalcoholic fatty liver disease: practice guidance from the American Association for the Study of Liver Diseases. Hepatology 2018;67: 328–357.
[46] Yoo ER, Kim D, Vazquez-Montesino LM, et al. Diet quality and its association with nonalcoholic fatty liver disease and all-cause and cause-specific mortality. Liver Int 2020;40:815–824.
[47] Chung GE, Youn J, Kim YS, et al. Dietary patterns are associated with the prevalence of nonalcoholic fatty liver disease in Korean adults. Nutrition 2019;62:32–38.
[48] Lang S, Martin A, Zhang X, et al. Combined analysis of gut microbiota, diet and PNPLA3 polymorphism in biopsy-proven non-alcoholic fatty liver disease. Liver Int 2021;41:1576–1591.
[49] Davis JN, Le KA, Walker RW, et al. Increased hepatic fat in overweight Hispanic youth influenced by interaction between genetic variation in PNPLA3 and high dietary carbohydrate and sugar consumption. Am J Clin Nutr 2010;92:1522–1527.
[50] Santoro N, Savoye M, Kim G, et al. Hepatic fat accumulation is modulated by the interaction between the rs738409 variant in the PNPLA3 gene and the dietary omega6/omega3 PUFA intake. PLoS One 2012;7: e37827.
[51] Nobili V, Liccardo D, Bedogni G, et al. Influence of dietary pattern, physical activity, and I148M PNPLA3 on steatosis severity in at-risk adolescents. Genes Nutr 2014;9:392.
[52] Vilar-Gomez E, Pirola CJ, Sookoian S, et al. Impact of the association between PNPLA3 genetic variation and dietary intake on the risk of significant fibrosis in patients with NAFLD. Am J Gastroenterol 2021;116: 994–1006.
[53] Kalafati IP, Dimitriou M, Borsa D, et al. Fish intake interacts with TM6SF2 gene variant to affect NAFLD risk: results of a case-control study. Eur J Nutr 2019;58:1463–1473.
[54] Wong VW, Wong GL, Chan RS, et al. Beneficial effects of lifestyle intervention in non-obese patients with non-alcoholic fatty liver disease. J Hepatol 2018;69:1349–1356.
[55] Katzmarzyk PT, Martin CK, Newton Jr RL, et al. Weight loss in underserved patients - a cluster-randomized trial. N Engl J Med 2020;383:909–918.
[56] Bradbury KE, Young HJ, Guo W, et al. Dietary assessment in UK Biobank: an evaluation of the performance of the touchscreen dietary questionnaire. J Nutr Sci 2018;7:e6.